

Customer Shopping Behavior Analysis

3,900 purchases — 18 columns — actionable insights





Dataset Summary

Rows

3,900

Columns

18 (demographics,
purchases, behavior)

Missing Data

37 values in Review Rating

- Data Prep
 - df.info(), .describe(), snake_case columns
- Missing Rating
 - Imputed by category median
- Feature Eng.
 - age_group, purchase_frequency_days
- DB
 - Loaded cleaned DataFrame to PostgreSQL

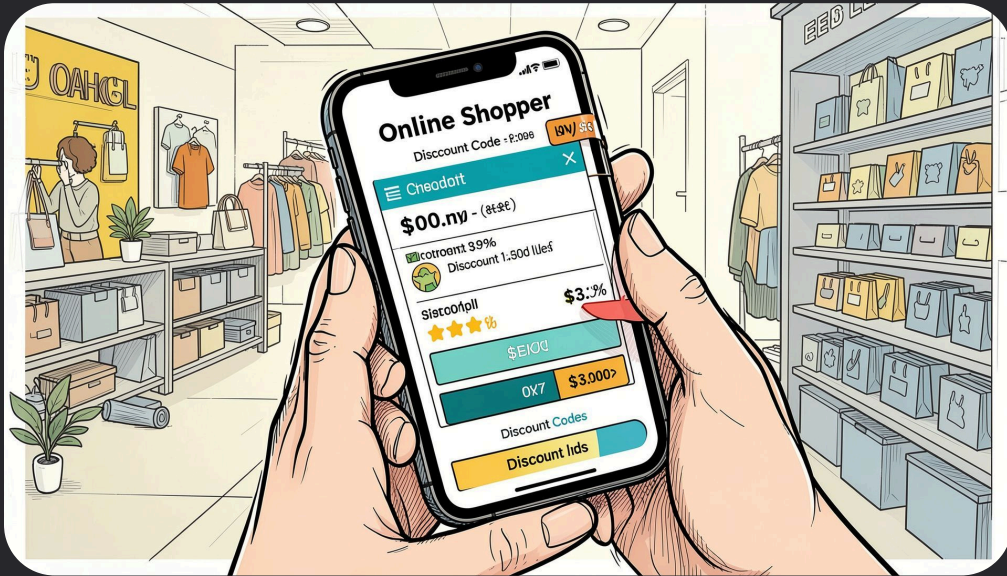




Revenue by Gender

Female: 75,191 — Male: 157,890

Discounts & High Spenders



High-Spending Discount Users

839 rows identified
(discount used &
above-average
spend)

Top Discounted Items

Hat 50%, Sneakers
49.66%, Coat 49.07%,
Sweater 48.17%,
Pants 47.37%

Top Products & Categories



Top Rated

Gloves 3.86, Sandals 3.84, Boots 3.82, Hat 3.80, Skirt 3.78



Top Sellers

Clothing: Blouse, Pants, Shirt

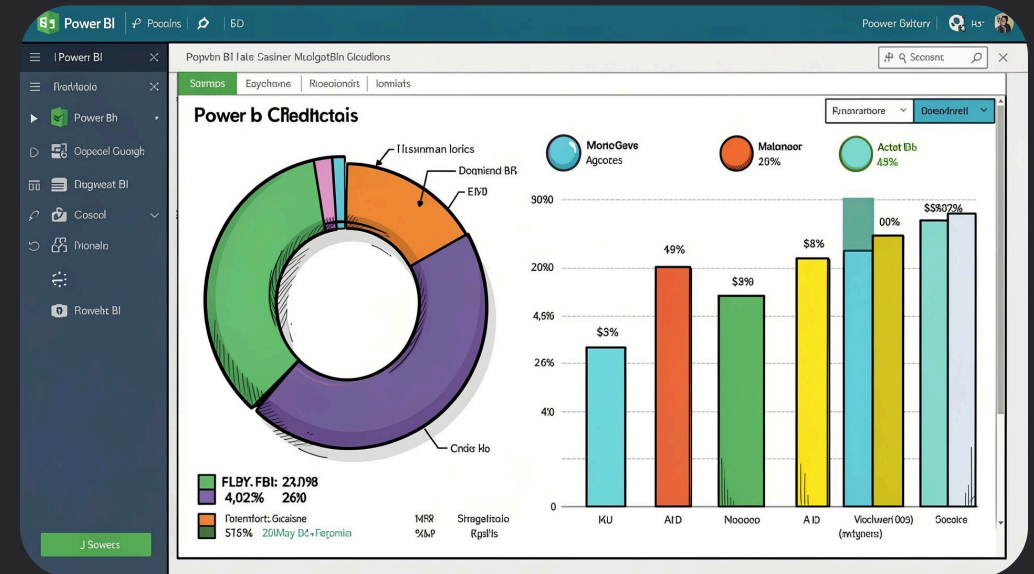
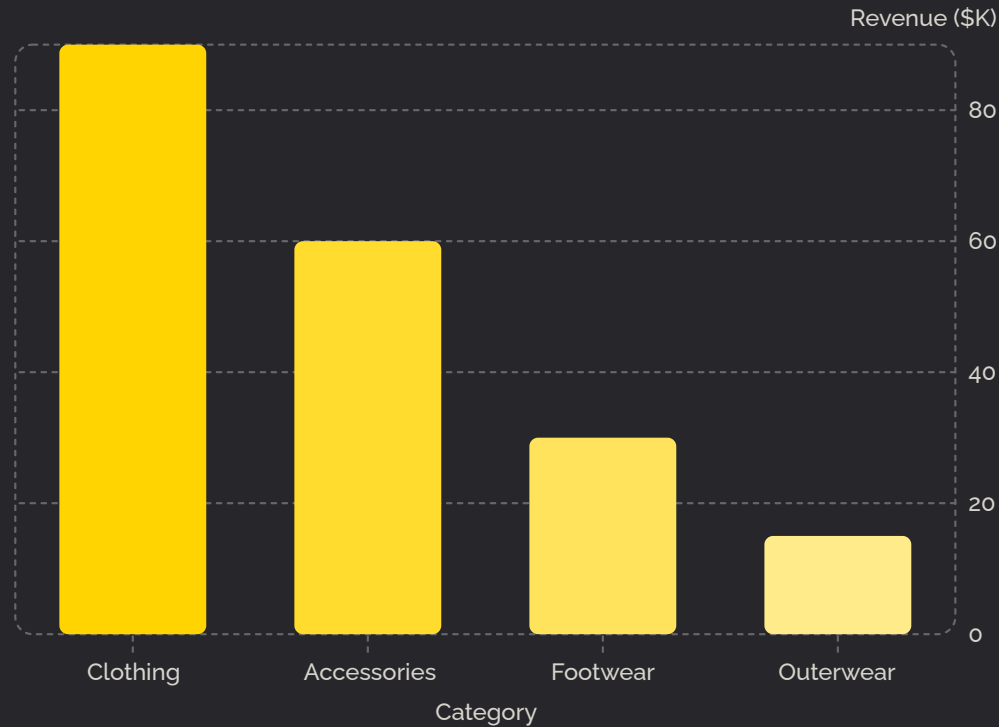


Accessories

Jewelry, Sunglasses, Belt (top orders)

Category Performance

Revenue by category (K): Clothing 90 · Accessories 60 · Footwear 30 · Outerwear 15



Audience & Spend



Segments

Loyal 3,116 · Returning 701 · New 83



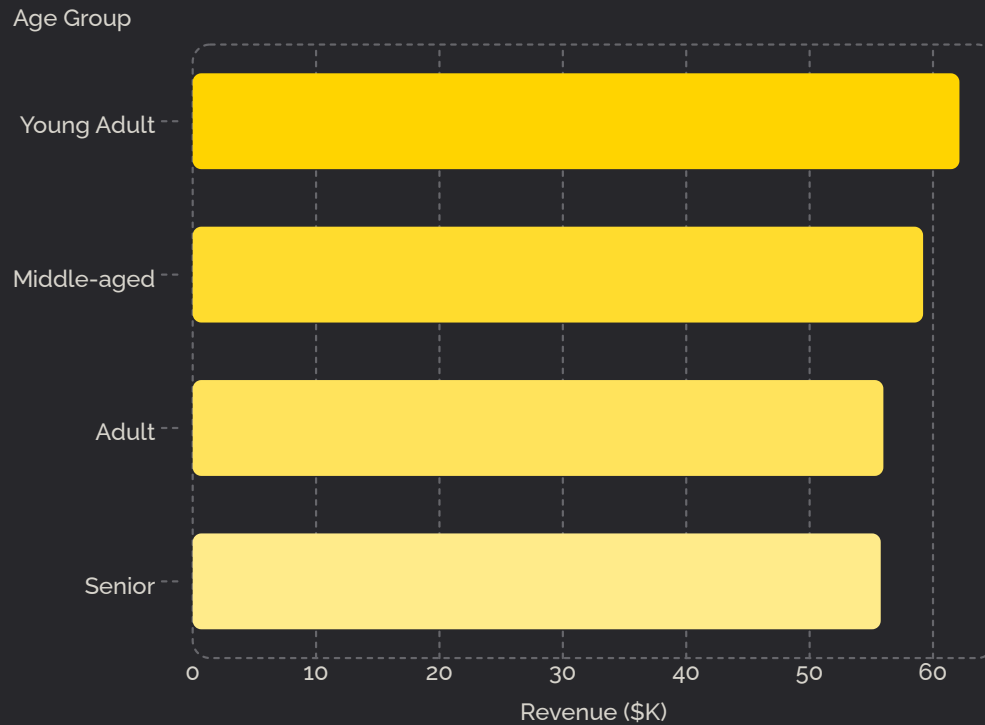
Shipping

Standard avg \$58.46 · Express avg \$60.48

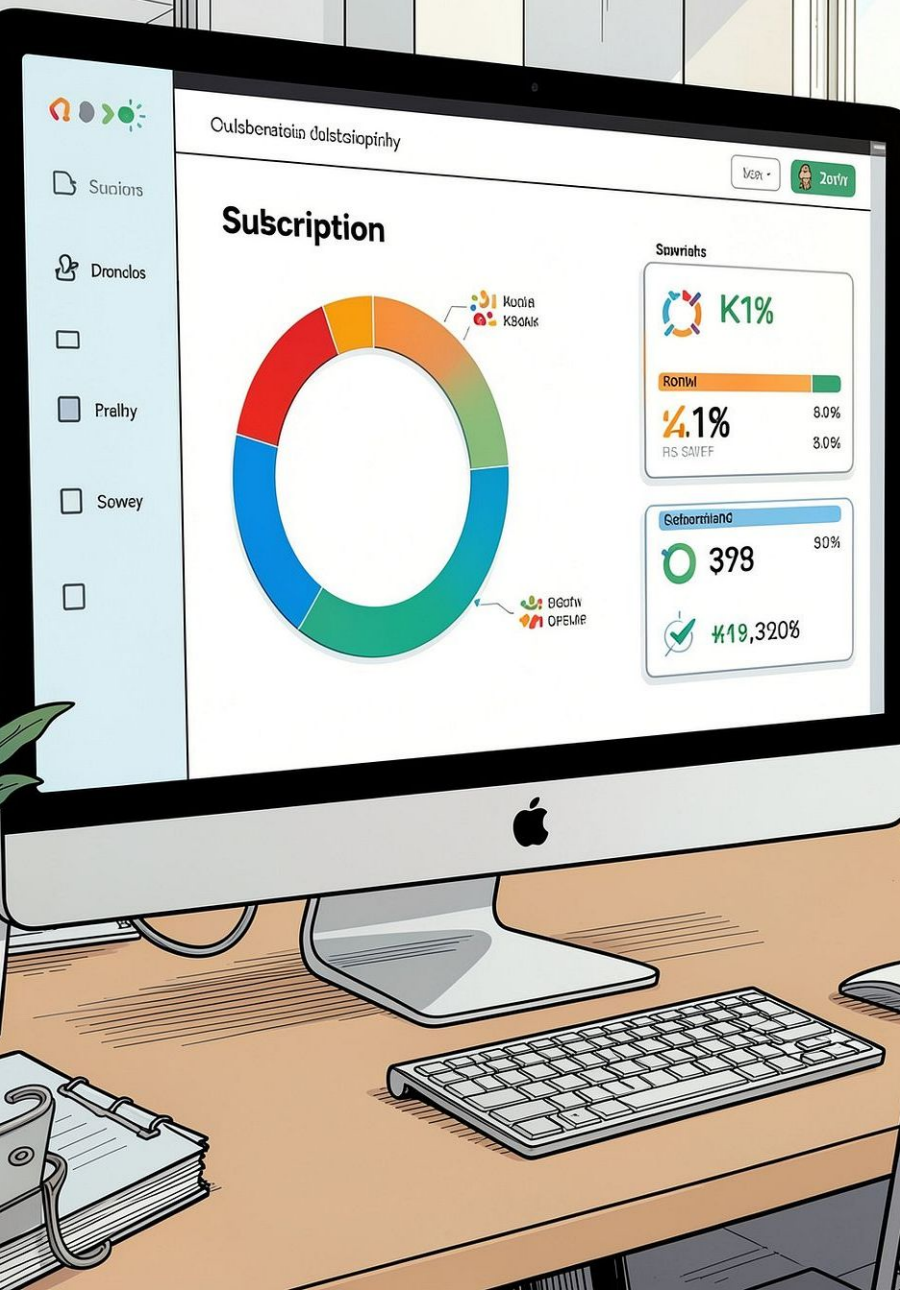


Payment

Most common: PayPal



Dashboard Highlights (Power BI)



Users

3.9K customers

Avg Spend

\$59.76

Avg Rating

3.75

Subscriptions

Yes 27% · No 73%

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to increase loyalty



Discount Policy

Balance promotions with margin control



Product Positioning

Feature top-rated & best-selling items



Targeted Marketing

Focus on high-revenue age groups & express-shipping users